

Module 4: Supervised Learning
Submodule: 4.6 Gaussian Discriminant Analysis
Lecture: Prediction with GDA

XCS229 Machine Learning

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I. Introduction

In this lecture, we discuss the concept of **prediction**. Our goal is to output a value y specifically to understand whether a given example x is benign cancer or not. (I.e., given x , output $y \in \{0, 1\}$.) The final goal is to determine whether the prediction for y indicates benign (0) or malignant cancer (1).

II. Maximizing Probability

To achieve this, we can state that we will output the most likely value of y . This is given by:

$$\arg \max p(y|x; \phi, \mu_0, \mu_1, \Sigma).$$

It is important to note that the parameters ϕ, μ_0, μ_1 and Σ are not arbitrary; they are solutions of MLE equations.

III. Maximizing Over Two Classes

In our case, we are maximizing over two possible choices for y : specifically whether the examples is more likely to be benign (0) or malignant (1). This leads us to compare two scalar values:

$$\arg \max \left\{ p(y = 1|x; \phi, \mu_0, \mu_1, \Sigma), p(y = 0|x; \phi, \mu_0, \mu_1, \Sigma) \right\}.$$

For simplicity, suppose:

$$\begin{aligned} A &= p(y = 1|x; \phi, \mu_0, \mu_1, \Sigma). \\ B &= p(y = 0|x; \phi, \mu_0, \mu_1, \Sigma). \end{aligned}$$

Thus, $A + B = 1$. Since we are interested in determining which of A or B is larger, we can redefine our problem. We are essentially checking if $A > 0.5$ or $A < 0.5$.

$$\max\{A, B\} = \begin{cases} A & \text{if } A \geq 0.5 \geq B, \\ B & \text{if } A \leq 0.5 \leq B. \end{cases}$$

So effectively we can summarize:

$$\max\{A, B\} = \begin{cases} 1 & \text{if } p(y = 1|x; \phi, \mu_0, \mu_1, \Sigma) \geq 0.5, \\ 0 & \text{if } p(y = 1|x; \phi, \mu_0, \mu_1, \Sigma) \leq 0.5. \end{cases}$$

IV. Decision Boundary

In this section, we explore the final decision and the boundary between two cases in terms of probability. The line that defines the decision boundary is characterized by the following condition:

$$\{x : p(y = 1|x) = 0.5\}$$

This means we are looking for a family of values x such that the probability of y given x equals exactly 0.5. This defines our decision boundary. On one side of this boundary, $p(y = 1|x)$ is more likely. Conversely, on the other side, $p(y = 0|x)$ is more likely. (It is essential to note that the boundary resulting from arbitrary groupings or random outputs will not frequently result in points lying precisely on the boundary.)

In the context of GDA, using Bayes' Rule we can reformulate our probability of interest:

$$p(y = 1|x) = \frac{p(x|y = 1)p(y = 1)}{p(x)}$$

Applying this to $p(y = 1|x; \phi, \mu_0, \mu_1, \Sigma)$ we get:

$$p(y = 1|x; \phi, \mu_0, \mu_1, \Sigma) = \frac{p(x | y = 1; \mu_1, \Sigma) \cdot p(y = 1; \phi)}{p(x; \phi, \mu_0, \mu_1, \Sigma)}.$$

For Gaussian Discriminant Analysis, if we view the quantity $p(y = 1|x; \phi, \mu_0, \mu_1, \Sigma)$ as a function of x , we find that it can be expressed in the form:

$$p(y = 1|x; \phi, \mu_0, \mu_1, \Sigma) = \frac{1}{1 + \exp(-(\theta^T x + \theta_0))}.$$

where $\theta_0 \in \mathbb{R}$ and $\theta \in \mathbb{R}^d$ is a function of the parameters $\phi, \mu_0, \mu_1, \Sigma$.

To find the decision boundary, we set the probability equal to 0.5:

$$\frac{1}{1 + \exp(-(\theta^T x + \theta_0))} = 0.5.$$

This simplifies to:

$$\exp(-(\theta^T x + \theta_0)) = 1.$$

Taking the natural logarithm gives:

$$-\theta^T x - \theta_0 = 0.$$

Which leads to:

$$\theta^T x + \theta_0 = 0.$$

The implication of this decision boundary is:

$$p(y = 1|x) > 0.5 \iff \theta^T x + \theta_0 > 0$$

This indicates that the decision boundary is a linear function of x , which separates the predicted classes effectively.

In summary, we determined that the decision boundary derived from GDA is linear and separates the two classes effectively, allowing us to classify cases of cancer as benign or malignant based on the features represented by x .

V. Summary

In this lecture, we covered essential concepts related to **Gaussian Discriminant Analysis** (GDA) and its application in prediction. Here are the key takeaways:

- **Objective of Prediction:** The primary goal is to classify a case as either benign cancer (0) or malignant cancer (1) based on a set of features represented by x .

- **Maximizing Probability:** We define the most likely value of y using:

$$\arg \max p(y|x; \phi, \mu_0, \mu_1, \Sigma).$$

- **Two Classes Comparison:** We compare probabilities for two classes, $A = p(y = 1|x)$ and $B = p(y = 0|x)$, leading to a decision based on whether $A > 0.5$ or $B > 0.5$.
- **Decision Boundary:** The line that separates the two classes is defined by the condition $\{x : p(y = 1|x) = 0.5\}$ and can be expressed as:

$$\theta^T x + \theta_0 = 0.$$

- **Bayes' Rule Application:** We utilize Bayes' Rule to derive the probability $p(y = 1|x)$, which leads us to the logistic function:

$$p(y = 1|x; \phi, \mu_0, \mu_1, \Sigma) = \frac{1}{1 + \exp(-(\theta^T x + \theta_0))}.$$

- **Linear Decision Boundary:** Finally, we conclude that the decision boundary is a linear function of x , which effectively separates benign from malignant cases.
- **Practical Implications:** Understanding this model equips us with the tools to classify cases in practical scenarios, emphasizing the importance of correctly estimating the parameters involved.